B Convex Optimization

In this appendix, we introduce the main definitions and results of convex optimization needed for the analysis of the learning algorithms presented in this book.

B.1 Differentiation and unconstrained optimization

We start with some basic definitions for differentiation needed to present Fermat's theorem and to describe some properties of convex functions.

Definition B.1 (Gradient) *Let $f\colon \mathcal{X} \subseteq \mathbb{R}^N \to \mathbb{R}$ be a differentiable function. Then, the* gradient *of f at $\mathbf{x} \in \mathcal{X}$ is the vector in $\mathbb{R}^N$ denoted by $\nabla f(\mathbf{x})$ and defined by*

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial \mathbf{x}_1}(\mathbf{x}) \\ \vdots \\ \frac{\partial f}{\partial \mathbf{x}_N}(\mathbf{x}) \end{bmatrix}.$$

Definition B.2 (Hessian) *Let $f\colon \mathcal{X} \subseteq \mathbb{R}^N \to \mathbb{R}$ be a twice differentiable function. Then, the* Hessian *of f at $\mathbf{x} \in \mathcal{X}$ is the matrix in $\mathbb{R}^{N \times N}$ denoted by $\nabla^2 f(\mathbf{x})$ and defined by*

$$\nabla^2 f(\mathbf{x}) = \Big[\frac{\partial^2 f}{\partial \mathbf{x}_i, \mathbf{x}_j}(\mathbf{x})\Big]_{1 \leq i,j \leq N}.$$

Next, we present a classic result for unconstrained optimization.

Theorem B.3 (Fermat's theorem) *Let $f\colon \mathcal{X} \subseteq \mathbb{R}^N \to \mathbb{R}$ be a differentiable function. If f admits a local extremum at $\mathbf{x}^* \in \mathcal{X}$, then $\nabla f(\mathbf{x}^*) = 0$, that is, $\mathbf{x}^*$ is a* stationary point.

B.2 Convexity

This section introduces the notions of *convex sets* and *convex functions*. Convex functions play an important role in the design and analysis of learning algorithms, in part because a local minimum of a convex function is necessarily also a global minimum. Thus, the properties of a hypothesis that is learned by finding a local minimum of a convex optimization are often well understood, while for some non-convex optimization problems there may be a very large number of local minima for which no clear characterization of the learned hypothesis can be given.

Definition B.4 (Convex set) *A set $\mathcal{X} \subseteq \mathbb{R}^N$ is said to be* convex *if for any two points $\mathbf{x}, \mathbf{y} \in \mathcal{X}$ the segment $[\mathbf{x}, \mathbf{y}]$ lies in $\mathcal{X}$, that is*

$$\{\alpha \mathbf{x} + (1-\alpha)\mathbf{y} \colon 0 \leq \alpha \leq 1\} \subseteq \mathcal{X}.$$

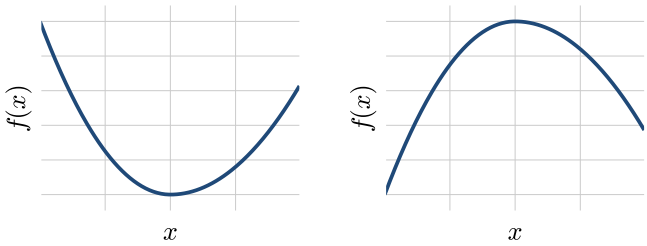

Figure B.1
Examples of a convex (left) and a concave (right) functions. Note that any line segment drawn between two points on the convex function lies entirely above the graph of the function while any line segment drawn between two points on the concave function lies entirely below the graph of the function.

The following lemma illustrates several operations on convex sets that preserve convexity. These will be useful for proving several subsequent results of this section.

Lemma B.5 (Operations that preserve convexity of sets) *The following operations on convex sets preserve convexity:*

- *Let $\{\mathcal{C}_i\}_{i\in I}$ be any family of sets where for all $i \in I$ the set $\mathcal{C}_i$ is convex. Then the intersection of these sets $\bigcap_{i\in I} \mathcal{C}_i$ is also convex.*
- *Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be convex sets, then their sum $\mathcal{C}_1 + \mathcal{C}_2 = \{x_1 + x_2 : x_1 \in \mathcal{C}_1, x_2 \in \mathcal{C}_2\}$, when defined, is convex.*
- *Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be convex sets, then their cross-product $(\mathcal{C}_1 \times \mathcal{C}_2)$ is also convex.*
- *Any projection of a convex set $\mathcal{C}$ is also convex.*

Proof: The first property holds since for any $x, y \in \bigcap_{i\in I} \mathcal{C}_i$ and any $\alpha \in [0, 1]$, we have $\alpha x + (1-\alpha)y \in \mathcal{C}_i$ for any $i \in I$ by the convexity of $\mathcal{C}_i$.

The second property holds since for any $(x_1 + x_2), (y_1 + y_2) \in (\mathcal{C}_1 + \mathcal{C}_2)$ we have $\alpha(x_1 + x_2) + (1-\alpha)(y_1 + y_2) = (\alpha x_1 + (1-\alpha)y_1 + \alpha x_2 + (1-\alpha)y_2) \in (\mathcal{C}_1 + \mathcal{C}_2)$, which follows since $\alpha x_1 + (1-\alpha)y_1 \in \mathcal{C}_1$ and $\alpha x_2 + (1-\alpha)y_2 \in \mathcal{C}_2$.

The third property holds since for $(x_1, x_2), (y_1, y_2) \in (\mathcal{C}_1 \times \mathcal{C}_2)$ we have $\alpha(x_1, x_2) + (1-\alpha)(y_1, y_2) = (\alpha x_1 + (1-\alpha)y_1, \alpha x_2 + (1-\alpha)y_2) \in (\mathcal{C}_1 \times \mathcal{C}_2)$, where the membership holds due to the assumption that $\mathcal{C}_1$ and $\mathcal{C}_2$ are convex.

Finally, the fourth property holds by noting that for any decomposition of the convex set $\mathcal{C}$ into projections $\mathcal{C}_1$ and $\mathcal{C}_2$, such that $\mathcal{C} = (\mathcal{C}_1 \times \mathcal{C}_2)$, it must be the case that $\mathcal{C}_1$ is convex. If $\mathcal{C}_2$ is empty, then the result is trivially true. Otherwise, fix an element $x_2 \in \mathcal{C}_2$, then for any $x, y \in \mathcal{C}_1$ and any $\alpha \in [0, 1]$ we have $\alpha(x, x_2) + (1-\alpha)(y, x_2) \in \mathcal{C}$, which implies $\alpha x + (1-\alpha)y \in \mathcal{C}_1$. Since $\mathcal{C}_1$ was chosen arbitrarily, this fact holds for any projection of $\mathcal{C}$. □

Note that many set operations may not preserve convexity. Consider the union of disjoint intervals on $\mathbb{R}$: $[a, b] \cup [c, d]$ where $a < b < c < d$. Clearly $[a, b]$ and $[c, d]$ are convex, however we have $\frac{1}{2}b + (1 - \frac{1}{2})c \notin ([a, b] \cup [c, d])$.

Definition B.6 (Convex hull) *The* convex hull $\mathrm{conv}(\mathcal{X})$ *of a set of points $\mathcal{X} \subseteq \mathbb{R}^N$ is the minimal convex set containing $\mathcal{X}$ and can be equivalently defined as follows:*

$$\mathrm{conv}(\mathcal{X}) = \Big\{ \sum_{i=1}^{m} \alpha_i \mathbf{x}_i : m \geq 1, \forall i \in [m], \mathbf{x}_i \in \mathcal{X}, \alpha_i \geq 0, \sum_{i=1}^{m} \alpha_i = 1 \Big\}. \tag{B.1}$$

Let $\mathrm{Epi}\, f$ denote the *epigraph* of function $f : \mathcal{X} \to \mathbb{R}$, that is the set of points lying above its graph: $\{(x, y) : x \in \mathcal{X}, y \geq f(x)\}$.

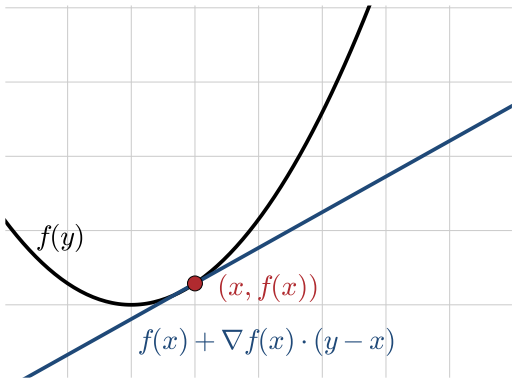

Figure B.2
Illustration of the first-order property satisfied by all convex functions.

Definition B.7 (Convex function) *Let $\mathcal{X}$ be a convex set. A function $f \colon \mathcal{X} \to \mathbb{R}$ is said to be* convex *iff* $\operatorname{Epi} f$ *is a convex set, or, equivalently, if for all $\mathbf{x}, \mathbf{y} \in \mathcal{X}$ and $\alpha \in [0, 1]$,*

$$f(\alpha \mathbf{x} + (1 - \alpha)\mathbf{y}) \leq \alpha f(\mathbf{x}) + (1 - \alpha) f(\mathbf{y}) \,. \tag{B.2}$$

f is said to be *strictly convex* if inequality (B.2) is strict for all $\mathbf{x}, \mathbf{y} \in \mathcal{X}$ where $\mathbf{x} \neq \mathbf{y}$ and $\alpha \in (0, 1)$. f is said to be (strictly) *concave* when $-f$ is (strictly) convex. Figure B.1 shows simple examples of convex and concave functions. Convex functions can also be characterized in terms of their first- or second-order differential.

Theorem B.8 *Let f be a differentiable function, then f is convex if and only if* $\operatorname{dom}(f)$ *is convex and the following inequalities hold:*

$$\forall \mathbf{x}, \mathbf{y} \in \operatorname{dom}(f), \;\; f(\mathbf{y}) - f(\mathbf{x}) \geq \nabla f(\mathbf{x}) \cdot (\mathbf{y} - \mathbf{x}) \,. \tag{B.3}$$

The property (B.3) is illustrated by figure B.2: for a convex function, the hyperplane tangent at $\mathbf{x}$ is always below the graph.

Theorem B.9 *Let f be a twice differentiable function, then f is convex iff* $\operatorname{dom}(f)$ *is convex and its Hessian is positive semidefinite:*

$$\forall \mathbf{x} \in \operatorname{dom}(f), \;\; \nabla^2 f(\mathbf{x}) \succeq 0 \,.$$

Recall that a symmetric matrix is positive semidefinite if all of its eigenvalues are non-negative. Further, note that when f is scalar, this theorem states that f is convex if and only if its second derivative is always non-negative, that is, for all $x \in \operatorname{dom}(f), f''(x) \geq 0$.

Example B.10 (Linear functions) Any linear function f is both convex and concave, since equation (B.2) holds with equality for both f and $-f$ by the definition of linearity.

Example B.11 (Quadratic function) The function $f \colon x \mapsto x^2$ defined over $\mathbb{R}$ is convex since it is twice differentiable and for all $x \in \mathbb{R}$, $f''(x) = 2 > 0$.

Example B.12 (Norms) Any norm $\|\cdot\|$ defined over a convex set $\mathcal{X}$ is convex since by the triangle inequality and the homogeneity property of the norm, for all $\alpha \in [0, 1], \mathbf{x}, \mathbf{y} \in \mathcal{X}$, we can write

$$\|\alpha \mathbf{x} + (1 - \alpha)\mathbf{y}\| \leq \|\alpha \mathbf{x}\| + \|(1 - \alpha)\mathbf{y}\| = \alpha \|\mathbf{x}\| + (1 - \alpha)\|\mathbf{y}\| \,.$$

Example B.13 (Maximum function) The max function defined for all $\mathbf{x} \in \mathbb{R}^N$, by $\mathbf{x} \mapsto \max_{j\in[N]} \mathbf{x}_j$ is convex. For all $\alpha \in [0,1], \mathbf{x}, \mathbf{y} \in \mathbb{R}^N$, by the sub-additivity of max, we can write

$$\max_j(\alpha\mathbf{x}_j + (1-\alpha)\mathbf{y}_j) \leq \max_j(\alpha\mathbf{x}_j) + \max_j((1-\alpha)\mathbf{y}_j) = \alpha \max_j(\mathbf{x}_j) + (1-\alpha)\max_j(\mathbf{y}_j)\,.$$

One useful approach for proving convexity or concavity of functions is to make use of composition rules. For simplicity of presentation, we will assume twice differentiability, although the results can also be proven without this assumption.

Lemma B.14 (Composition of convex/concave functions) *Assume $h : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R}^N \to \mathbb{R}$ are twice differentiable functions and for all $\mathbf{x} \in \mathbb{R}^N$, define $f(\mathbf{x}) = h(g(\mathbf{x}))$. Then the following implications are valid:*

- *h is convex and non-decreasing, and g is convex $\implies$ f is convex.*
- *h is convex and non-increasing, and g is concave $\implies$ f is convex.*
- *h is concave and non-decreasing, and g is concave $\implies$ f is concave.*
- *h is concave and non-increasing, and g is convex $\implies$ f is concave.*

Proof: We restrict ourselves to $N = 1$, since it suffices to prove convexity (concavity) along all arbitrary lines that intersect the domain. Now, consider the second derivative of f:

$$f''(x) = h''(g(x))g'(x)^2 + h'(g(x))g''(x)\,. \tag{B.4}$$

Note that if h is convex and non-decreasing, we have $h'' \geq 0$ and $h' \geq 0$. Furthermore, if g is convex we also have $g'' \geq 0$, and it follows that $f''(x) \geq 0$, which proves the first statement. The remainder of the statements are proven in a similar manner. □

Example B.15 (Composition of functions) The previous lemma shows the convexity or concavity of the following composed functions:

- If $f : \mathbb{R}^N \to \mathbb{R}$ is convex, then $\exp(f)$ is convex.
- Any squared norm $\|\cdot\|^2$ is convex.
- For all $\mathbf{x} \in \mathbb{R}^N$ the function $\mathbf{x} \mapsto \log(\sum_{j=1}^N x_j)$ is concave.

The following two lemmas give examples of two other operations preserving convexity.

Lemma B.16 (Pointwise supremum or maximum of convex functions) *Let $(f_i)_{i\in\mathcal{I}}$ be a family of convex functions defined over a convex set $\mathcal{C}$. Then, their* pointwise supremum *f defined for all $x \in \mathcal{C}$ by $f(x) = \sup_{i\in\mathcal{I}} f_i(x)$ (resp. their* pointwise maximum *if $|\mathcal{I}| < +\infty$) is a convex function.*

Proof: Observe that $\operatorname{Epi} f = \cap_{i\in\mathcal{I}} \operatorname{Epi} f_i$ and is therefore convex as an intersection of convex sets. □

Example B.17 (Pointwise supremum of convex functions) The lemma shows in particular the convexity of the following functions:

- A piecewise linear function f defined for all $x \in \mathbb{R}^N$ by $f(x) = \max_{i\in[m]} \mathbf{w}_i^\top \mathbf{x} + b_i$ is convex as a pointwise maximum of affine (and thus convex) functions.
- The maximum eigenvalue $\lambda_{\max}(\mathbf{M})$ is a convex function over the set of symmetric matrices $\mathbf{M}$ since the set of symmetric matrices is convex and since $\lambda_{\max}(\mathbf{M}) = \sup_{\|\mathbf{x}\|_2\leq 1} \mathbf{x}^\top\mathbf{M}\mathbf{x}$ is defined as the supremum of the linear (and thus convex) functions $\mathbf{M} \mapsto \mathbf{x}^\top\mathbf{M}\mathbf{x}$.
- More generally, let $\lambda_1(\mathbf{M}), \ldots, \lambda_k(\mathbf{M})$ denote the top $k \leq n$ eigenvalues of a symmetric $n \times n$ matrix $\mathbf{M}$. Then, by a similar argument, $\mathbf{M} \mapsto \sum_{i=1}^k \lambda_i(\mathbf{M})$ is a convex function since $\sum_{i=1}^k \lambda_i(\mathbf{M}) = \sup_{\dim(\mathbf{V})=k} \sum_{i=1}^k \mathbf{u}_i^\top\mathbf{M}\mathbf{u}_i$, where $\mathbf{u}_1, \ldots, \mathbf{u}_k$ is an orthonormal basis of $\mathbf{V}$.
- Using the previous property, along with the fact that $\operatorname{Tr}(\mathbf{M})$ is linear in $\mathbf{M}$, also shows that $\mathbf{M} \mapsto \sum_{i=k+1}^n \lambda_i(\mathbf{M}) = \operatorname{Tr}(\mathbf{M}) - \sum_{i=1}^k \lambda_i(\mathbf{M})$ or $\mathbf{M} \mapsto \sum_{i=n-k+1}^n \lambda_i(\mathbf{M}) = -\sum_{i=1}^k \lambda_i(-\mathbf{M})$ are concave functions.

Lemma B.18 (Partial infimum) *Let f be a convex function defined over a convex set $\mathcal{C} \subseteq \mathcal{X} \times \mathcal{Y}$ and let $\mathcal{B} \subseteq \mathcal{Y}$ be a convex set such that $\mathcal{A} = \{x \in \mathcal{X} \colon \exists y \in \mathcal{B} \mid (x, y) \in \mathcal{C}\}$ is non-empty. Then, $\mathcal{A}$ is a convex set and the function g defined for all $x \in \mathcal{A}$ by $g(x) = \inf_{y \in \mathcal{B}} f(x, y)$ is convex.*

Proof: First note that the intersection of the convex sets $\mathcal{C}$ and $(\mathcal{X} \times \mathcal{B})$ is convex. Thus, $\mathcal{A}$ is convex since it is the projection of the convex set $\mathcal{C} \cap (\mathcal{X} \times \mathcal{B})$ onto $\mathcal{X}$.

Let x_1 and x_2 be in $\mathcal{A}$. By definition of g, for any $\epsilon > 0$, there exist $y_1, y_2 \in \mathcal{B}$ with $(x_1, y_1), (x_2, y_2) \in \mathcal{C}$ such that $f(x_1, y_1) \leq g(x_1) + \epsilon$ and $f(x_2, y_2) \leq g(x_2) + \epsilon$. Then, for any $\alpha \in [0, 1]$,

$$\begin{aligned} g(\alpha x_1 + (1-\alpha)x_2) &= \inf_{y \in \mathcal{B}} f(\alpha x_1 + (1-\alpha)x_2, y) \\ &\leq f(\alpha x_1 + (1-\alpha)x_2, \alpha y_1 + (1-\alpha)y_2) \\ &\leq \alpha f(x_1, y_1) + (1-\alpha) f(x_2, y_2) \\ &\leq \alpha g(x_1) + (1-\alpha) g(x_2) + \epsilon. \end{aligned}$$

Since the inequality holds for all $\epsilon > 0$, it implies

$$g(\alpha x_1 + (1-\alpha)x_2) \leq \alpha g(x_1) + (1-\alpha) g(x_2),$$

which completes the proof. □

Example B.19 The lemma shows in particular that the distance to a convex set $\mathcal{B}$, $d(x, \mathcal{B}) = \inf_{y \in \mathcal{B}} \|x - y\|$, is a convex function of x in any normed vector space, since $(x, y) \mapsto \|x - y\|$ is jointly convex in x and y for any norm $\|\cdot\|$.

The following is a useful inequality applied in a variety of contexts. It is in fact a quasi-direct consequence of the definition of convexity.

Theorem B.20 (Jensen's inequality) *Let X be a random variable taking values in a non-empty convex set $\mathcal{C} \subseteq \mathbb{R}^N$ with a finite expectation $\mathbb{E}[X]$, and f a measurable convex function defined over $\mathcal{C}$. Then, $\mathbb{E}[X]$ is in $\mathcal{C}$, $\mathbb{E}[f(X)]$ is finite, and the following inequality holds:*

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)].$$

Proof: We give a sketch of the proof, which essentially follows from the definition of convexity. Note that for any finite set of elements $x_1, \ldots, x_n$ in $\mathcal{C}$ and any positive reals $\alpha_1, \ldots, \alpha_n$ such that $\sum_{i=1}^n \alpha_i = 1$, we have

$$f\Big(\sum_{i=1}^n \alpha_i x_i\Big) \leq \sum_{i=1}^n \alpha_i f(x_i)\,.$$

This follows straightforwardly by induction from the definition of convexity. Since the α_is can be interpreted as probabilities, this immediately proves the inequality for any distribution with a finite support defined by $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_n)$:

$$f(\mathop{\mathbb{E}}_{\boldsymbol{\alpha}}[X]) \leq \mathop{\mathbb{E}}_{\boldsymbol{\alpha}}[f(X)]\,.$$

Extending this to arbitrary distributions can be shown via the continuity of f on any open set, which is guaranteed by the convexity of f, and the weak density of distributions with finite support in the family of all probability measures. □

B.3 Constrained optimization

We now define a general constrained optimization problem and the specific properties associated to convex constrained optimization problems.

Definition B.21 (Constrained optimization problem) *Let $\mathcal{X} \subseteq \mathbb{R}^N$ and $f, g_i \colon \mathcal{X} \to \mathbb{R}$, for all $i \in [m]$. Then, a* constrained optimization problem *has the form:*

$$\begin{aligned} &\min_{\mathbf{x} \in \mathcal{X}} \quad f(\mathbf{x}) \\ &\textit{subject to:} \quad g_i(\mathbf{x}) \leq 0, \ \forall i \in \{1, \ldots, m\}. \end{aligned}$$

This general formulation does not make any convexity assumptions and can be augmented with equality constraints. It is referred to as the *primal problem* in contrast with a related problem introduced later. We will denote by p^* the optimal value of the objective.

For any $\mathbf{x} \in \mathcal{X}$, we will denote by $g(\mathbf{x})$ the vector $(g_1(\mathbf{x}), \ldots, g_m(\mathbf{x}))^\top$. Thus, the constraints can be written as $g(\mathbf{x}) \leq \mathbf{0}$. To any constrained optimization problem, we can associate a *Lagrange function* that plays an important role in the analysis of the problem and its relationship with another related optimization problem.

Definition B.22 (Lagrangian) *The* Lagrange function *or the* Lagrangian *associated to the general constrained optimization problem defined in (B.21) is the function defined over $\mathcal{X} \times \mathbb{R}_+$ by:*

$$\forall \mathbf{x} \in \mathcal{X}, \forall \boldsymbol{\alpha} \geq 0, \quad \mathcal{L}(\mathbf{x}, \boldsymbol{\alpha}) = f(\mathbf{x}) + \sum_{i=1}^m \alpha_i g_i(\mathbf{x}),$$

where the variables α_i are known as the Lagrange *or* dual *variables with $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_m)^\top$.*

Any equality constraint of the form $g(\mathbf{x}) = 0$ for a function g can be equivalently expressed by two inequalities: $-g(\mathbf{x}) \leq 0$ and $+g(\mathbf{x}) \leq 0$. Let $\alpha_- \geq 0$ be the Lagrange variable associated to the first constraint and $\alpha_+ \geq 0$ the one associated to the second constraint. The sum of the terms corresponding to these constraints in the definition of the Lagrange function can therefore be written as $\alpha g(\mathbf{x})$ with $\alpha = (\alpha_+ - \alpha_-)$. Thus, in general, for an equality constraint $g(\mathbf{x}) = 0$ the Lagrangian is augmented with a term $\alpha g(\mathbf{x})$ but with $\alpha \in \mathbb{R}$ not constrained to be non-negative. Note that in the case of a convex optimization problem, equality constraints $g(\mathbf{x})$ are required to be affine since both $g(\mathbf{x})$ and $-g(\mathbf{x})$ are required to be convex.

Definition B.23 (Dual function) *The* (Lagrange) dual function *associated to the constrained optimization problem is defined by*

$$\forall \boldsymbol{\alpha} \geq 0, F(\boldsymbol{\alpha}) = \inf_{\mathbf{x} \in \mathcal{X}} \mathcal{L}(\mathbf{x}, \boldsymbol{\alpha}) = \inf_{\mathbf{x} \in \mathcal{X}} \left(f(\mathbf{x}) + \sum_{i=1}^m \alpha_i g_i(\mathbf{x}) \right). \tag{B.5}$$

Note that F is always concave, since the Lagrangian is linear with respect to $\boldsymbol{\alpha}$ and since the infimum preserves concavity. We further observe that

$$\forall \boldsymbol{\alpha} \geq 0, \quad F(\boldsymbol{\alpha}) \leq p^*, \tag{B.6}$$

since for any feasible $\mathbf{x}$, $f(\mathbf{x}) + \sum_{i=1}^m \alpha_i g_i(\mathbf{x}) \leq f(\mathbf{x})$. The dual function naturally leads to the following optimization problem.

Definition B.24 (Dual problem) *The* dual (optimization) problem *associated to the constrained optimization problem is*

$$\begin{aligned} &\max_{\boldsymbol{\alpha}} \quad F(\boldsymbol{\alpha}) \\ &\textit{subject to:} \quad \boldsymbol{\alpha} \geq 0. \end{aligned}$$

The dual problem is always a convex optimization problem (as a maximization of a concave problem). Let d^* denote an optimal value. By (B.6), the following inequality always holds:

$$d^* \leq p^* \qquad (\textit{weak duality}).$$

The difference $(p^* - d^*)$ is known as the *duality gap*. The equality case

$$d^* = p^* \qquad (\textit{strong duality})$$

does not hold in general. However, strong duality does hold when convex problems satisfy a *constraint qualification*. We will denote by $\mathrm{int}(\mathcal{X})$ the interior of the set $\mathcal{X}$.

Definition B.25 (Strong constraint qualification) *Assume that* $\mathrm{int}(\mathcal{X}) \neq \emptyset$. *Then, the* strong constraint qualification *or* Slater's condition *is defined as*

$$\exists\, \overline{\mathbf{x}} \in \mathrm{int}(\mathcal{X})\colon g(\overline{\mathbf{x}}) < 0. \tag{B.7}$$

A function $h\colon \mathcal{X} \to \mathbb{R}$ is said to be *affine* if it can be defined for all $\mathbf{x} \in \mathcal{X}$ by $h(\mathbf{x}) = \mathbf{w} \cdot \mathbf{x} + b$, for some $\mathbf{w} \in \mathbb{R}^N$ and $b \in \mathbb{R}$.

Definition B.26 (Weak constraint qualification) *Assume that* $\mathrm{int}(\mathcal{X}) \neq \emptyset$. *Then, the* weak constraint qualification *or* weak Slater's condition *is defined as*

$$\exists\, \overline{\mathbf{x}} \in \mathrm{int}(\mathcal{X})\colon \forall i \in [m], \big(g_i(\overline{\mathbf{x}}) < 0\big) \vee \big(g_i(\overline{\mathbf{x}}) = 0 \wedge g_i \ \textit{affine}\big). \tag{B.8}$$

We next present sufficient and necessary conditions for solutions to constrained optimization problems, based on the saddle point of the Lagrangian and Slater's condition.

Theorem B.27 (Saddle point — sufficient condition) *Let P be a constrained optimization problem over $\mathcal{X} = \mathbb{R}^N$. If $(\mathbf{x}^*, \boldsymbol{\alpha}^*)$ is a saddle point of the associated Lagrangian, that is,*

$$\forall \mathbf{x} \in \mathbb{R}^N, \forall \boldsymbol{\alpha} \geq 0, \quad \mathcal{L}(\mathbf{x}^*, \boldsymbol{\alpha}) \leq \mathcal{L}(\mathbf{x}^*, \boldsymbol{\alpha}^*) \leq \mathcal{L}(\mathbf{x}, \boldsymbol{\alpha}^*), \tag{B.9}$$

then $\mathbf{x}^*$ is a solution of the problem P.

Proof: By the first inequality, the following holds:

$$\begin{aligned}\forall \boldsymbol{\alpha} \geq 0, \mathcal{L}(\mathbf{x}^*, \boldsymbol{\alpha}) \leq \mathcal{L}(\mathbf{x}^*, \boldsymbol{\alpha}^*) &\Rightarrow \forall \boldsymbol{\alpha} \geq 0, \boldsymbol{\alpha} \cdot g(\mathbf{x}^*) \leq \boldsymbol{\alpha}^* \cdot g(\mathbf{x}^*) \\ &\Rightarrow g(\mathbf{x}^*) \leq 0 \wedge \boldsymbol{\alpha}^* \cdot g(\mathbf{x}^*) = 0\,, \end{aligned} \tag{B.10}$$

where $g(\mathbf{x}^*) \leq 0$ in (B.10) follows by letting $\boldsymbol{\alpha} \to +\infty$ and $\boldsymbol{\alpha}^* \cdot g(\mathbf{x}^*) = 0$ follows by letting $\boldsymbol{\alpha} \to 0$. In view of (B.10), the second inequality in (B.9) gives,

$$\forall \mathbf{x}, \mathcal{L}(\mathbf{x}^*, \boldsymbol{\alpha}^*) \leq \mathcal{L}(\mathbf{x}, \boldsymbol{\alpha}^*) \Rightarrow \forall \mathbf{x}, f(\mathbf{x}^*) \leq f(\mathbf{x}) + \boldsymbol{\alpha}^* \cdot g(\mathbf{x}).$$

Thus, for all $\mathbf{x}$ satisfying the constraints, that is $g(\mathbf{x}) \leq 0$, we have

$$f(\mathbf{x}^*) \leq f(\mathbf{x}),$$

which completes the proof. □

Theorem B.28 (Saddle point — necessary condition) *Assume that f and g_i, $i \in [m]$, are* convex functions *and that Slater's condition holds. Then, if $\mathbf{x}$ is a solution of the constrained optimization problem, there exists $\boldsymbol{\alpha} \geq 0$ such that $(\mathbf{x}, \boldsymbol{\alpha})$ is a saddle point of the Lagrangian.*

Theorem B.29 (Saddle point — necessary condition) *Assume that f and g_i, $i \in [m]$, are* convex differentiable functions *and that the weak Slater's condition holds. If $\mathbf{x}$ is a solution of the constrained optimization problem, then there exists $\boldsymbol{\alpha} \geq 0$ such that $(\mathbf{x}, \boldsymbol{\alpha})$ is a saddle point of the Lagrangian.*

We conclude with a theorem providing necessary and sufficient optimality conditions when the problem is convex, the objective function differentiable, and the constraints qualified.

Theorem B.30 (Karush-Kuhn-Tucker's theorem) *Assume that $f, g_i\colon \mathcal{X} \to \mathbb{R}, \forall i \in [m]$ are convex and differentiable and that the constraints are qualified. Then $\overline{\mathbf{x}}$ is a solution of the constrained program if and if only there exists $\overline{\boldsymbol{\alpha}} \geq 0$ such that,*

$$\nabla_{\mathbf{x}} \mathcal{L}(\overline{\mathbf{x}}, \overline{\boldsymbol{\alpha}}) = \nabla_{\mathbf{x}} f(\overline{\mathbf{x}}) + \overline{\boldsymbol{\alpha}} \cdot \nabla_{\mathbf{x}} g(\overline{\mathbf{x}}) = 0 \tag{B.11}$$

$$\nabla_{\boldsymbol{\alpha}} \mathcal{L}(\overline{\mathbf{x}}, \overline{\boldsymbol{\alpha}}) = g(\overline{\mathbf{x}}) \leq 0 \tag{B.12}$$

$$\overline{\boldsymbol{\alpha}} \cdot g(\overline{\mathbf{x}}) = \sum_{i=1}^{m} \overline{\boldsymbol{\alpha}}_i g_i(\overline{\mathbf{x}}) = 0\,. \tag{B.13}$$

The conditions B.11–B.13 are known as the KKT conditions. *Note that the last two KKT conditions are equivalent to*

$$g(\overline{\mathbf{x}}) \leq 0 \wedge (\forall i \in \{1, \ldots, m\}, \bar{\alpha}_i g_i(\overline{\mathbf{x}}) = 0). \tag{B.14}$$

These equalities are known as complementarity conditions.

Proof: For the forward direction, since the constraints are qualified, if $\overline{\mathbf{x}}$ is a solution, then there exists $\overline{\boldsymbol{\alpha}}$ such that the $(\overline{\mathbf{x}}, \overline{\boldsymbol{\alpha}})$ is a saddle point of the Lagrangian and all three conditions are satisfied (the first condition follows by definition of a saddle point, and the second two conditions follow from (B.10)).

In the opposite direction, if the conditions are met, then for any $\mathbf{x}$ such that $g(\mathbf{x}) \leq 0$, we can write

$$\begin{aligned} f(\mathbf{x}) - f(\overline{\mathbf{x}}) &\geq \nabla_{\mathbf{x}} f(\overline{\mathbf{x}}) \cdot (\mathbf{x} - \overline{\mathbf{x}}) && \text{(convexity of } f\text{)} \\ &= -\sum_{i=1}^{m} \overline{\boldsymbol{\alpha}}_i \nabla_{\mathbf{x}} g_i(\overline{\mathbf{x}}) \cdot (\mathbf{x} - \overline{\mathbf{x}}) && \text{(first condition)} \\ &\geq -\sum_{i=1}^{m} \overline{\boldsymbol{\alpha}}_i [g_i(\mathbf{x}) - g_i(\overline{\mathbf{x}})] && \text{(convexity of } g_i\text{s)} \\ &= -\sum_{i=1}^{m} \overline{\boldsymbol{\alpha}}_i g_i(\mathbf{x}) \geq 0, && \text{(third condition)} \end{aligned}$$

which shows that $f(\overline{\mathbf{x}})$ is the minimum of f over the set of points satisfying the constraints. □

B.4 Fenchel duality

In this section, we present an alternative theory of convex optimization or convex analysis where the functions f considered may be non-differentiable and take infinite values.

Throughout, this section, the set $\mathcal{X}$ denotes a Hilbert space with the inner product denoted by $\langle \cdot, \cdot \rangle$. However, the results presented can be straightforwardly extended to the case of a Banach space. We consider functions taking values in $[-\infty, +\infty]$. The *domain of a function* $f \colon \mathcal{X} \to [-\infty, +\infty]$ is defined as the set

$$\mathrm{dom}(f) = \{x \in \mathcal{X} \colon f(x) < +\infty\}. \tag{B.15}$$

We extend the definition of convexity and say that $f \colon \mathcal{X} \to [-\infty, +\infty]$ is convex if it is convex over $\mathrm{dom}(f)$, that is if for all $x, x' \in \mathrm{dom}(f)$ and all $t \in [0, 1]$,

$$f(tx + (1-t)x') \leq tu + (1-t)v, \tag{B.16}$$

for all $(u, v) \in \mathbb{R}^2$ with $u \geq f(x)$ and $v \geq f(x')$. A convex function is said to be *proper* if it takes values in $(-\infty, +\infty]$ and if it is not uniformly equal to $+\infty$. It is said to be *closed* when its epigraph is closed.

B.4.1 Subgradients

Definition B.31 *Let $f \colon \mathcal{X} \to (-\infty, +\infty]$ be a convex function. Then, a vector $g \in \mathcal{X}$ is a* subgradient *of f at a point $x \in \mathrm{dom}(f)$ if the following inequality holds for all $z \in \mathcal{X}$:*

$$f(z) \geq f(x) + \langle z - x, g \rangle. \tag{B.17}$$

The set of all subgradients at x is called the subdifferential of f at x *and is denoted by $\partial f(x)$ with $\partial f(x) = \emptyset$ for $x \notin \mathrm{dom}(f)$.*

Thus, g is a subgradient at x iff the hyperplane with normal vector g passing through the point $(x, f(x))$ is below the graph of f, that is iff it is supporting the graph of f. Figure 14.1 illustrates these definitions.

The following lemma shows that if f is differentiable at $x \in \mathrm{dom}(f)$, then its subdifferential is reduced to its gradient at x.

Lemma B.32 *If f is differentiable at $x \in \mathrm{dom}(f)$, then $\partial f(x) = \{\nabla f(x)\}$.*

Proof: Clearly, the gradient $\nabla f(x)$ is always a subgradient at x. Now, let g be in $\partial f(x)$. Then, by definition of a subgradient, for any $\epsilon \in \mathbb{R}$,

$$f(x + \epsilon(\nabla f(x) - g)) \geq f(x) + \epsilon\langle \nabla f(x) - g, g\rangle.$$

A first-order Taylor series expansion gives

$$f(x + \epsilon(\nabla f(x) - g)) - f(x) = \epsilon\langle \nabla f(x), \nabla f(x) - g\rangle + o(\epsilon\|\nabla f(x) - g\|).$$

In view of that, the first inequality can be rewritten as

$$\epsilon\|\nabla f(x) - g\|^2 \leq o(\epsilon\|\nabla f(x) - g\|),$$

which implies $\|\nabla f(x) - g\| = o(1)$ and $\nabla f(x) = g$. □

Proposition B.33 *Let $f\colon \mathcal{X} \to (-\infty, +\infty]$ be a proper function. Then, x^* is a global minimizer of f iff $\partial f(x^*)$ contains* 0.

Proof: Since f is proper, if x^* is a minimizer, $f(x^*)$ cannot be $+\infty$. Thus x^* must be in $\mathrm{dom}(f)$ (and thus $\partial f(x)$ is not defined to be empty). Now, x^* is a global minimizer iff for all $z \in \mathcal{X}$, $f(z) \geq f(x^*)$, that is iff 0 is a subgradient of f at x^*. □

B.4.2 Core

The *core* of a set $\mathcal{C} \subseteq \mathcal{X}$ is denoted by $\mathrm{core}(\mathcal{C})$ and defined as follows:

$$\mathrm{core}(\mathcal{C}) = \{x \in \mathcal{C}\colon \forall u \in \mathcal{X}, \exists \epsilon > 0 \mid \forall t \in [0, \epsilon], (x + tu) \in \mathcal{C}\}. \tag{B.18}$$

Thus, for $x \in \mathrm{core}(\mathcal{C})$, for any direction u, $(x + tu)$ is in $\mathcal{C}$ for t sufficiently small. In view of this definition, $\mathrm{core}(\mathcal{C})$ clearly contains the interior of $\mathcal{C}$, $\mathrm{int}(\mathcal{C})$.

Proposition B.34 *Let $h\colon \mathcal{X} \to [-\infty, +\infty]$ be a convex function. If there exists $x_0 \in \mathrm{core}(\mathrm{dom}(h))$ such that $h(x_0) > -\infty$, then $h(x) > -\infty$ for all $x \in \mathcal{X}$.*

Proof: Let x be in $\mathcal{X}$. Since x_0 is in $\mathrm{core}(\mathrm{dom}(h))$, there exists $t > 0$ such that $x_0' = x_0 + t(x_0 - x)$ is in $\mathrm{dom}(h)$, that is such that $h(x_0') < +\infty$. Since $x_0 = \frac{1}{1+t}x_0' + \frac{t}{1+t}x$, by convexity, the following holds:

$$h(x_0) \leq \frac{1}{1+t}h(x_0') + \frac{t}{1+t}v \iff v \geq \frac{1+t}{t}\Big[h(x_0) - \frac{1}{1+t}h(x_0')\Big] \tag{B.19}$$

for all $v \geq h(x)$. This implies $h(x) \geq \frac{1+t}{t}[h(x_0) - \frac{1}{1+t}h(x_0')] > -\infty$, which concludes the proof. □

The proof of the following result is left as an exercise (Exercise B.3).

Proposition B.35 *Let $h\colon \mathcal{X} \to (-\infty, +\infty]$ be a convex function. Then, h admits a subdifferential at any $x \in \mathrm{core}(\mathrm{dom}(h))$.*

B.4.3 Conjugate functions

Definition B.36 (Conjugate functions) *Let $\mathcal{X}$ be a Hilbert space. The* conjugate function *or* Fenchel conjugate *of a function $f\colon \mathcal{X} \mapsto [-\infty, +\infty]$ is the function $f^*\colon \mathcal{X} \mapsto [-\infty, +\infty]$ defined by*

$$f^*(u) = \sup_{x \in \mathcal{X}} \big\{\langle u, x\rangle - f(x)\big\}. \tag{B.20}$$

Note that f^* is convex as the pointwise supremum of the set of affine and thus convex functions $u \mapsto \langle x, u\rangle - f(x)$. Also, if there exists x such that $f(x) < +\infty$, then $f > -\infty$. Conjugation is order-reversing: for any f and g, if $f \leq g$, then $g^* \leq f^*$. Also, it is straightforward to see that if f is closed proper convex, then $f^{**} = f$.

Figure B.3 illustrates the definition of conjugate functions. As shown by the figure, conjugate functions correspond to a dual description of the epigraph of a function in terms of supporting hyperplanes and their crossing points.

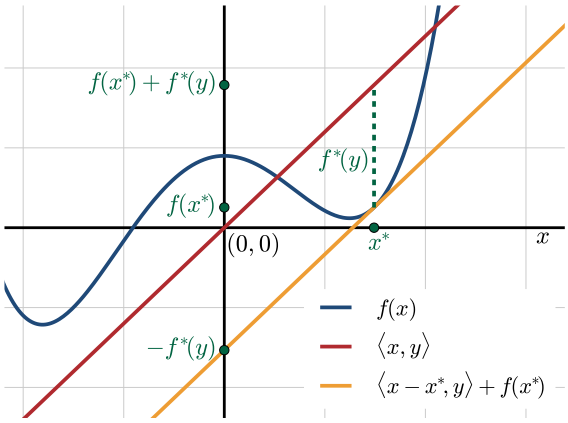

Figure B.3
Illustration of the conjugate f^* of a function f. Given y, x^* is the point at which the distance between the hyperplane of equation $z = \langle x, y\rangle$ with normal y (slope y in dimension one) and the plot of $f(x)$ is the largest. This largest distance is equal to $f^*(y)$. The parallel hyperplane $z = \langle x - x^*, y\rangle + f(x^*)$ with normal y and passing through the point $(x^*, f(x^*)$ is shown. This is a supporting hyperplane of the plot of $f(x)$. The point at which it intercepts the y-axis (crossing point) has y-coordinate $-f^*(y)$.

Lemma B.37 (Conjugate of extended relative entropy) *Let $p_0 \in \Delta$ be a distribution over $\mathcal{X}$ such that $p_0(x) > 0$ for all $x \in \mathcal{X}$. Define $f\colon \mathbb{R}^{\mathcal{X}} \to \mathbb{R}$ by*

$$f(p) = \begin{cases} \mathsf{D}(p\|p_0) & \text{if } p \in \Delta \\ +\infty & \text{otherwise.} \end{cases}$$

Then, the conjugate function $f^*\colon \mathbb{R}^{\mathcal{X}} \to \mathbb{R}$ of f is defined by

$$\forall q \in \mathbb{R}^{\mathcal{X}},\ f^*(q) = \log\left(\sum_{x\in\mathcal{X}} p_0(x)e^{q(x)}\right).$$

Proof: By definition of f, for any $q \in \mathbb{R}^{\mathcal{X}}$, we can write

$$\sup_{p\in\mathbb{R}^{\mathcal{X}}} \big(\langle p, q\rangle - \mathsf{D}(p\|p_0)\big) = \sup_{p\in\Delta} \big(\langle p, q\rangle - \mathsf{D}(p\|p_0)\big). \tag{B.21}$$

Fix $q \in \mathbb{R}^{\mathcal{X}}$ and let $\bar{q} \in \Delta$ be defined for all $x \in \mathcal{X}$ by

$$\bar{q}(x) = \frac{p_0(x)e^{q(x)}}{\sum_{x\in\mathcal{X}} p_0(x)e^{q(x)}} = \frac{p_0(x)e^{q(x)}}{\mathbb{E}_{p_0}[e^q]}. \tag{B.22}$$

Then, the following holds for all $p \in \Delta$:

$$\langle p, q\rangle - \mathsf{D}(p\|p_0) = \mathop{\mathbb{E}}_{p}[\log(e^q)] - \mathop{\mathbb{E}}_{p}\left[\log\frac{p}{p_0}\right] = \mathop{\mathbb{E}}_{p}\left[\log\frac{p_0 e^q}{p}\right] = -\mathsf{D}(p\|\bar{q}) + \log \mathop{\mathbb{E}}_{p_0}[e^q].$$

Since $\mathsf{D}(p\|\bar{q}) \geq 0$ and $\mathsf{D}(p\|\bar{q}) = 0$ for $p = \bar{q}$, this shows that $\sup_{p\in\Delta}\big(p\cdot q - \mathsf{D}(p\|p_0)\big) = \log\big(\mathbb{E}_{p_0}[e^q]\big)$ and concludes the proof. □

Table B.1 gives a series of other examples of functions and their conjugates. The following is an immediate consequence of the definition of the conjugate functions.

Table B.1
Examples of functions g and their conjugates g^*.

$g(x)$	$\mathrm{dom}(g)$	$g^*(y)$	$\mathrm{dom}(g^*)$
$f(ax)\ (a \neq 0)$	$\mathcal{X}$	$f^*\left(\frac{y}{a}\right)$	$\mathcal{X}^*$
$f(x+b)$	$\mathcal{X}$	$f^*(y) - \langle b, y\rangle$	$\mathcal{X}^*$
$af(x)\ (a > 0)$	$\mathcal{X}$	$af^*\left(\frac{y}{a}\right)$	$\mathcal{X}^*$
$\alpha x + \beta$	$\mathbb{R}$	$\begin{cases} -\beta & \text{if } y = \alpha \\ +\infty & \text{otherwise} \end{cases}$	$\mathbb{R}$
$\frac{\lvert x\rvert^p}{p}\ (p > 1)$	$\mathbb{R}$	$\frac{\lvert y\rvert^q}{q}\ (\frac{1}{p} + \frac{1}{q} = 1)$	$\mathbb{R}$
$\frac{-x^p}{p}\ (0 < p < 1)$	$\mathbb{R}_+$	$\frac{-(-y)^q}{q}\ (\frac{1}{p} + \frac{1}{q} = 1)$	$\mathbb{R}_-$
$\sqrt{1+x^2}$	$\mathbb{R}$	$-\sqrt{1-(y)^2}$	$[-1, 1]$
$-\log(x)$	$\mathbb{R}_+ \setminus \{0\}$	$-(1+\log(-y))$	$\mathbb{R}_- \setminus \{0\}$
e^x	$\mathbb{R}$	$\begin{cases} y\log(y) - y\,, & \text{if } y > 0 \\ 0\,, & \text{if } y = 0 \end{cases}$	$\mathbb{R}_+$
$\log(1+e^x)$	$\mathbb{R}$	$\begin{cases} y\log(y) + (1-y)\log(1-y)\,, & \text{if } 0 < y < 1 \\ 0\,, & \text{if } y = 0, 1 \end{cases}$	$[0, 1]$
$-\log(1-e^x)$	$\mathbb{R}$	$\begin{cases} y\log(y) - (1+y)\log(1+y)\,, & \text{if } y > 0 \\ 0\,, & \text{if } y = 0 \end{cases}$	$\mathbb{R}_+$

Proposition B.38 (Fenchel's inequality) *Let $\mathcal{X}$ be a Hilbert space. For any function $f\colon \mathcal{X} \mapsto [-\infty, +\infty]$ and any $x \in \mathrm{dom}(f)$ and $u \in \mathcal{X}$, the following inequality holds:*

$$f(x) + f^*(u) \geq \langle u, x\rangle\,. \tag{B.23}$$

Equality holds iff u is a subgradient of f at x.

We will denote by A^* the adjoint operator of a bounded (or continuous) linear map $A\colon \mathcal{X} \to \mathcal{Y}$. Also, we denote by $\mathrm{cont}(f)$ the set of points $x \in \mathcal{X}$ at which $f\colon \mathcal{X} \to [-\infty, +\infty]$ is finite and continuous.

Theorem B.39 (Fenchel duality theorem) *Let $\mathcal{X}$ and $\mathcal{Y}$ be two Hilbert spaces, $f\colon \mathcal{X} \to (-\infty, +\infty]$ and $g\colon \mathcal{Y} \to (-\infty, +\infty]$ two convex functions and $A\colon \mathcal{X} \to \mathcal{Y}$ a bounded linear map. Then, the following two optimization problems (*Fenchel problems*)*

$$p^* = \inf_{x \in \mathcal{X}} \{f(x) + g(Ax)\}$$
$$d^* = \sup_{y \in \mathcal{Y}} \{-f^*(A^*y) - g^*(-y)\}$$

satisfy the weak duality $p^* \geq d^*$. *If further f and g satisfy the condition*

$$0 \in \mathrm{core}\big(\mathrm{dom}(g) - A(\mathrm{dom}(f))\big),$$

or the stronger condition

$$A(\mathrm{dom}(f)) \cap \mathrm{cont}(g) \neq \emptyset,$$

then strong duality *holds, that is $p^* = d^*$ and the supremum in the dual problem is attained if $d^* \in \mathbb{R}$.*

Proof: By Fenchel's inequality (proposition B.38) applied to both f and g, for any $x \in \mathcal{X}$ and $y \in \mathcal{Y}$, the following inequalities hold:

$$f(x) + f^*(A^*y) \geq \langle A^*y, x\rangle = \langle y, Ax\rangle = -\langle -y, Ax\rangle \geq -g(Ax) - g^*(-y).$$

Comparing the leftmost and the rightmost terms gives

$$f(x) + f^*(A^*y) \geq -g(Ax) - g^*(-y) \iff f(x) + g(Ax) \geq -f^*(A^*y) - g^*(-y).$$

Taking the infimum over $x \in \mathcal{X}$ of the left-hand side and the supremum over $y \in \mathcal{Y}$ of the right-hand side of the last inequality yields $p^* \geq d^*$.

Consider now the function $h\colon \mathcal{Y} \to [-\infty, +\infty]$ defined for all $u \in \mathcal{Y}$ by

$$h(u) = \inf_{x\in\mathcal{X}} \{f(x) + g(Ax + u)\}. \tag{B.24}$$

Since $(x, u) \mapsto f(x) + g(Ax + u)$ is convex, h is convex as the infimum over one argument of that function. u is in $\operatorname{dom}(h)$ iff there exists $x \in \mathcal{X}$ such that $f(x) + g(Ax + u) < +\infty$, that is iff there exists $x \in \mathcal{X}$ such that $f(x) < +\infty$ and $g(Ax + u) < +\infty$, that is iff there exists $x \in \operatorname{dom}(f)$ such that $(Ax + u) \in \operatorname{dom}(g)$. Thus, we have $\operatorname{dom}(h) = \operatorname{dom}(g) - A\operatorname{dom}(f)$.

If $p^* = -\infty$, then strong duality clearly holds. Otherwise, $p^* > -\infty$. If $0 \in \operatorname{core}\big(\operatorname{dom}(g) - A(\operatorname{dom}(f))\big) = \operatorname{core}(\operatorname{dom}(h))$, then 0 is in $\operatorname{dom}(h)$ and $p^* < +\infty$. Thus, $p^* = h(0)$ is in $\mathbb{R}$. By proposition B.34, since $h(0) > -\infty$ and $0 \in \operatorname{core}(\operatorname{dom}(h))$, h takes values in $(-\infty, +\infty]$. Thus, by proposition B.35, h admits a subgradient $-y$ at 0. By definition of y, for all $x \in \mathcal{X}$ and $u \in \mathcal{Y}$,

$$\begin{aligned}
h(0) &\leq h(u) + \langle y, u\rangle \\
&\leq f(x) + g(Ax + u) + \langle y, u\rangle \\
&= \{f(x) - \langle A^*y, x\rangle\} + \{g(Ax + u) + \langle y, u\rangle + \langle A^*y, x\rangle\} \\
&= \{f(x) - \langle A^*y, x\rangle\} + \{g(Ax + u) + \langle y, Ax + u\rangle\}.
\end{aligned}$$

Taking the infimum over u and the supremum over x yields

$$h(0) \leq -f^*(A^*y) - g^*(-y) \leq d^* \leq p^* = h(0),$$

which proves $d^* = p^*$ and that the supremum defining d^* is reached at y.

Finally, assume that $A(\operatorname{dom}(f)) \cap \operatorname{cont}(g) \neq \emptyset$ and let $u \in A(\operatorname{dom}(f)) \cap \operatorname{cont}(g)$. Then, $u = Ax$ with $x \in \operatorname{dom}(f)$ and $u \in \operatorname{cont}(g) \subsetneq \operatorname{dom}(g)$. Thus, we have $0 = u - Ax \in \operatorname{dom}(g) - A\operatorname{dom}(f)$. Since g is continuous at u and $g(u)$ is finite, for any $v \in \mathcal{X}$, there exists $\epsilon > 0$ such that $g(u + tv)$ is finite for all $t \in [0, \epsilon]$, thus $w_t = (u + tv) \in \operatorname{dom}(g)$. Therefore, for any $t \in [0, \epsilon]$, $tv = w_t - u = w_t - Ax \in \operatorname{dom}(g) - A\operatorname{dom}(f)$, which shows that $0 \in \operatorname{core}\big(\operatorname{dom}(g) - A(\operatorname{dom}(f))\big)$. □

To illustrate the theorem, consider the case where A is the identity operator. The primal optimization problem is then $\min_x \big\{f(x) + g(x)\big\}$. Figure B.4 illustrates the Fenchel duality theorem in that case. The primal problem consists of finding the point x^* at which the distance between the plots of $f(x)$ and $-g(x)$ is minimal since $f(x) + g(x) = f(x) - (-g(x))$. As shown by the figure, under the conditions of the theorem, this coincides with seeking y^* at which the difference of the conjugate values of $f(x)$ and $-g(x)$, that is the difference $-f^*(y) - g^*(-y)$ is maximal.

B.5 Chapter notes

The results presented in this appendix are based on several key theorems: theorem B.3 due to Fermat (1629); theorem B.27 due to Lagrange (1797), theorem B.30 due to Karush [1939] and Kuhn and Tucker [1951], and theorem B.39 due to Werner Fenchel, based on the notion of conjugate functions or Legendre transformations. For a more extensive material on convex optimization, we strongly recommend the books of Boyd and Vandenberghe [2004], Bertsekas, Nedić, and Ozdaglar [2003], Rockafellar [1997], Borwein and Lewis [2000] and Borwein and Zhu [2005] which have formed the basis for part of the material presented in this appendix. In particular, our table of conjugate functions is extracted from [Borwein and Lewis, 2000].

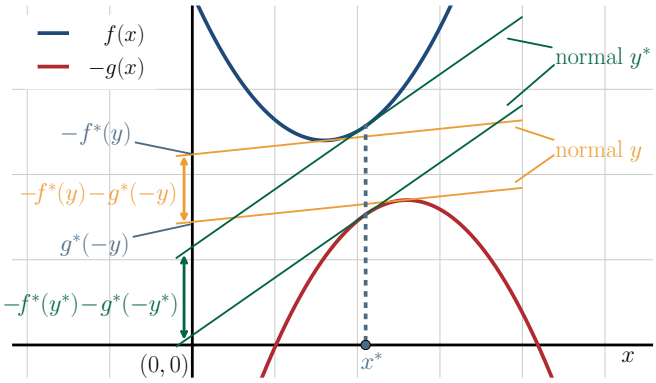

Figure B.4
Illustration of Fenchel Duality; $\min_x \{f(x)+g(x)\} = \max_y \{-f^*(y)-g^*(-y)\}$.

B.6 Exercises

B.1 Give the conjugate of the function f defined by $f(x) = |x|$ for all $x \in \mathbb{R}$.

B.2 Prove the correctness of the conjugate function g^* for each function g of Table B.1.

B.3 Give the proof of Proposition B.35.